Mandy Wang

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EDUCATION

New York University, Tandon School of Engineering Sep 2024 - May 2028
B.S. Computer Science | GPA: 3.68
Relevant Coursework: Design & Analysis of Algorithms, Data Structures, Object Oriented Programming, Databases, Computer Architecture

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, HTML, CSS, JavaScript, SQL, Verilog
Frameworks & Libraries: ROS2, OpenCV, PyTorch, pygame, OpenAI Whisper
Hardware & Tools: Git, Linux, Jetson Orin Nano, Raspberry Pi, Arduino, RealSense, Gazebo

PROJECTS

Voice-Interactive Autonomous Mobile Robot May 2026 - Present

- Debugged STM32F446RE motor firmware for PWM control, encoder feedback, and PID speed regulation, and integrated it with the Raspberry Pi over UART for motor commands and telemetry
- Developed wheel odometry from encoder data and converted Nav2 velocity commands into wheel speeds using differential-drive kinematics, fusing odometry with an Extended Kalman Filter
- Wrote a ROS2 driver for a time-of-flight depth camera and converted its depth images to laser scans for slam_toolbox mapping, configured Nav2 for autonomous navigation
- Developing a voice interaction system that transcribes speech with OpenAI Whisper, classifies user intent, and displays pygame animation on a monitor screen

Music Vibration Vest Jan 2025 - May 2025

- Programmed a wearable that translates audio into haptic feedback for people with hearing impairments through vibration motors
- Built a real-time Python signal processing pipeline that detects beats in bass, mid, and treble bands using Butterworth bandpass filters and Hilbert envelopes, scaling motor intensity by FFT band energy
- Helped program Arduino firmware that parses Bluetooth serial commands and control vibration motors with PWM duty cycles mapped to each band's intensity

EXPERIENCE

Software Lead, **NYU Robotics Design Team -** *New York, NY* Sep 2025 - Present

- Lead a 10+ member software subteam developing a ROS2 autonomy stack for a NASA Lunabotics excavation rover, spanning Nav2, RTAB-Map SLAM with the RealSense D455, AprilTag localization, and mission state machines
- Coordinate with mechanical and electrical subteams as a systems engineer through system design reviews to define rover requirements, strategy, and hardware-software integration with the Jetson Orin Nano and Teensy
- Developed the initial AprilTag localization pipeline with apriltag_ros, calibrating onboard webcams, validating multi-tag detection and chaining camera, tag, and robot TF transforms to estimate the rover's pose on the map

Course Assistant, **NYU Tandon Computer Science Department -** *New York, NY* Sep 2025 - Present

- Lead weekly lab sections for 50+ students, teaching core Python concepts and guiding them through coding problems
- Hold weekly one-on-one office hours to help students debug and troubleshoot their Python code
- Collaborate with professors to grade homework assignments and lab exams

Student Researcher, **NYU Flexible AI-Enabled Mechatronics Systems Lab (FAMS) -** *New York, NY* Sep 2025 - Jan 2026

- Worked on the computer vision subteam to prototype a 3D CNN in PyTorch and MONAI that classifies kidney tumors in KiTS23 CT scans
- Generated Grad-CAM heatmaps to visualize which regions of each scan influenced the model's predictions